Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Agrawal, Ajay K. Ph.D.		
Project Title:	Gravitational Effects on Flow Instability and Transition in Low Density Jets		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	35487	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	04/01/2000	**End Date:**	08/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	7	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Griffin, DeVon	**Contact Phone:**	216-433-8109
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Flight Program:			
Flight Assignment:	note: extended 8 months		
Key Personnel Changes/Previous PI:	Agrawal, Ajay		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2388		
Performance Goal No.:			
Performance Goal Text:			

The primary objective of this research is to quantify how buoyancy affects flow instability and structure in the near field of low-density jets. In particular, we are interested in studying low-density jets that display self-excited behavior on Earth. By quantifying the flow field in both normal and reduced gravity, we plan to understand the fundamental basis for this instability.

This research has three scientific objectives: (1) investigate the effects of gravity on flow instability and structure of low-density, buoyant jets (in normal gravity) that display self-excited behavior. By removing buoyancy in our experiments, we seek to obtain direct physical evidence of the instability mechanism. (2) determine if the instability in momentum-driven, low-density jets is buoyancy driven. By eliminating buoyancy in microgravity environment, globally as well as locally, we seek to examine our hypothesis that the instability mechanism in self-excited momentum-driven

Task Description:	driven jets is buoyancy driven. (3) perform linear stability analysis of low-density gas jets with the inclusion of gravitational effects. We seek to study the characteristics of both convective and absolute instabilities in low-density jets and demarcate the boundary separating the two instability modes. Low-density jets occur in exhaust from engines and stacks in the industry, geophysical events such as volcanic eruptions, fuel leaks, diffusion flames, and accidental fires. The jet flows are also important in extraterrestrial applications, for example, in space propulsion, materials processing in space and other planets. The fluid dynamic instabilities in the shear layer between the jet and ambient fluid, the formation and characteristics of large-scale coherent structures, entrainment of the ambient fluid into the jet flow, and the mean flow field, which affect the evolution and transport of the structures are important near-field phenomena. The near-field jet behavior has direct influence on the acoustic noise and flow development in the far-field. The study closes major gaps in understanding of how buoyancy affects the flow instability and transition in low-density jets. The study will determine if buoyancy effects on the flow instability can be characterized using global parameters.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress provided this reporting period.
Bibliography Type:	Description: (Last Updated: 02/19/2019)
Articles in Peer-reviewed Journals	Pasumarthi KS, Agrawal AK. "Schlieren measurements and analysis of concentration field in self-excited helium jets." Physics of Fluids. 2003 Dec;15(12):3683. https://doi.org/10.1063/1.1623490 , Dec-2003
Articles in Peer-reviewed Journals	Yep T-W, Agrawal AK, Griffin D. "Gravitational effects on near field flow structure of low-density gas jets." AIAA Journal. 2003 Oct;41(10):1973-9. https://doi.org/10.2514/2.1887 , Oct-2003